Théo Lisart

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www.linkedin.com/in/theo-lisart
github.com/tlisart (in construction)

Education

Master of Applied Science | ***Physics engineering : Nuclear and quantum engineering*** — Sept. 2019 – May 2021
École Polytechnique — Brussels, Belgium

Bachelor of Applied Science | ***Major: Engineering, Minor: Physics engineering*** — Sept. 2014 – May 2019
École Polytechnique — Brussels, Belgium

Cambridge english certification (CEA) — Sep. 2013 – May 2014
EF language school — Vancouver, BC

Secondary Education — Sep. 2007 – May 2013
Collège St. Gertrude — Nivelles, Belgium

Work and Experience

Master thesis — September 2020 – now (August 2021)
QuiC departement at ULB — Brussels, Belgium

- **On building a computational model of Markovian adiabatic quantum computation (working title)**
  The current paradigm of circuit-based quantum computation has seen tremendous practical progress in the last 10 years. Adiabatic quantum computing has been theoretically explored however lacks a complete computational framework empowering researchers to discover new algorithms. We hope to explore the effects of decoherence on driven Hamiltonian, and if it can be advantageous for computing.

Association co-creator : Cit'light (continuous learning) — 2019 – now
In charge of — Brussels, Belgium

- Writing and article correction
- Expantion strategy and reach-out about humain-technology relations
- Technology and democracy/urbanism
- Participative project management (OPAL)

Micro-electronics manual work — 2020
Shakmat modular synthesis — Brussels, Belgium

- Soldering and packaging products

Touring guide and educator — 2015-2016
Expérimentarium de l'ULB — Brussels, Belgium

- Teaching 9-12 y.o. children the scientific method : empiricism and hypothetico -deduction concepts with small and fun experiments.
- Vibrations and light demonstrations for 13-17 y.o. adolescents

Academic Projects (selected)

FabZero ULB program — 2021
Fablab ULB | *Numerical manufacturing, multi-disciplinary*

- Learning through frugal design project digital fabrication : 3D printing, laser and vinyl cutting, applied electronics and PCB soldering

Using IA to improve in-core fuel management of PWR nuclear reactors — 2020
ENGIE-Tractebel | *BASH, Python, research*

- Establish a state-of-the-art and strategies in the use of Machine learning in the design process of in-core loading patterns

Using Machine Learning for Rumor detection 2020

ML/datascience course | *Python*

- Using text pattern recognition methods, building a micro-blogging rumor detection model based on Twitter and Weibo data.
- Complete quality analysis on LSTM, GRU and usual RNN

Diffusion-convection partial differential equation (multigrid) 2018-2019

Numerical analysis I and II | *C*

- Homogeneous PDE solver (time-evolution, eigenvalue problem)
- Non-homogeneous BC : multigrid solver from scratch
- Convergence analysis

Modal analysis of lower vibrating modes on structure 2020

Vibrations and Acoustics | *MATLAB*

- Extraction of physical data from hammer modal test
- FRF, H-estimator, modal parameter validation

Wi-Fi ray-tracing application 2019

Telecommunications physics | *C++*

- Recursive ray-tracing algorithms (Transmission/reflection/diffraction)
- GUI in Qt framework, power-heatmapping of reception in designable rooms

"Mushroom" role playing game 2016

Object oriented programming | *JAVA, UML*

- Role playing game using model-view-controller design
- Animations, serialization, encapsulation, maze-generator, complete concept

About me

Generalities

- Physics engineering student passionate about a large range of topics. My main focus is on quantum theory and its technological applications (telecom, cryptography, computing, sensing and control), particularly interested in the deeper ontological difficulties of the field.
- I believe that software is art so is the act of building new technologies.
- I am interested in management, work better with people than alone, as it is with interactions that great ideas can emerge, I am generally the creative motor of the team.
- My side-subjects are political philosophy, associative work and music.
- I practice swimming and thai boxing.

Advocacy

Right to repair, free-software and Opensource, high-tech/low-tech, free access to educative content
internet security, collaborative democracy, fablab philosophy

Personal Projects (selected)

The MATH experimental sound suite (not ready for release, nor branding) | ***C++*** 2020

VST3 standard digital instruments and effects for experimenting with math and physics on MIDI or soundstreams.
Higher-dimensionnal algebra and soundesign experiments,
physical modeling synthesis (digital waveguide, etc)
Machine Learning sound synthesis
complex plane, fractals and sound.

Helping hand at Fablab autonomous effort for coronavirus protection | **/** Summer 2020

Contributed to the spontaneous effort from ULB to make protective gears in the fabrication lab, small help in low-tech design for use by non-expert manufacturer volunteers.

Skills

Languages: French (Native), English (C1-advance)
IT: Linux (fluent), BASH and basis in netsec and infosec
Programming: Efficient in C, C++, Python(Numpy, Pandas, TF/Keras, IBM-Qskit),
Document Creation: Libre Office Suite, LaTex
Content creation: Soundesign in Ableton or Bitwig
Miscellaneous: Red-cross first aid certificate